



df Command in Linux

Introduction

Imagine you need to:

- Check available disk space on your system.
- Monitor storage usage before installing large files.
- Find out which partition is filling up quickly.

How do you check disk space efficiently?

This is where the **df (disk free) command** helps!

Section 1: Learn

What is **df**?

The **df** command in Linux displays disk space usage for each file system. It helps:

1. Monitor available and used storage on all mounted partitions.
2. Identify which disk is running out of space.
3. Check the percentage of disk usage.

Why Do We Need **df**?

Without df	With df
No way to check available disk space	Displays free and used disk space
System may run out of storage unexpectedly	Helps prevent storage-related crashes
Hard to track partition usage	Shows usage per partition



Without df	With df
No storage optimization insights	Helps manage storage efficiently

How Does **df** Work?

1. It scans all mounted file systems.
2. It calculates used, free, and total disk space.
3. It presents the results in a readable format.

An Interesting Anecdote

Before tools like **df**, system admins had to manually calculate disk space by summing file sizes! The **df** command was introduced to **automate this process and prevent storage failures**.

Section 2: Practice

Step-by-Step Guide to Using **df**

1. Checking Basic Disk Usage

```
# Display disk space usage for all file systems
df
```

What happens here?

- Lists **total, used, and available disk space**.
 - Shows **usage in kilobytes by default**.
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2. Displaying Human-Readable Format (-h)

```
# Show disk space in MB/GB instead of bytes  
df -h
```

Why is this useful?

- Easier to **read and analyze storage values**.
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3. Checking Usage of a Specific Partition

```
# Show disk space for root (`/') partition only  
df -h /
```

How does this help?

- Focuses **only on relevant partitions**.
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4. Displaying File System Type (-T)

```
# Show disk usage along with file system type  
df -T
```

Colum n	Meaning
Filesystem	Name of the mounted file system
Type	File system type (e.g., ext4, xfs)
Size	Total disk space
Used	Space currently used



Colum n	Meaning
Availab le	Free space left
Use%	Percentage of used space
Mounte d on	Mount point (where the file system is attached)

Why is this important?

- Helps identify **file system types and storage allocation**.
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5. Checking Storage in a Specific Unit (-BM, -BG)

```
# Show disk space in MB
```

```
df -BM
```

```
# Show disk space in GB
```

```
df -BG
```

What does this do?

- Displays **storage in megabytes or gigabytes**, making it easier to analyze.
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6. Showing Inodes Instead of Storage (-i)

```
# Display inode usage instead of storage space
```

```
df -i
```

Why is this useful?



- **Inodes track files and directories**; if they run out, no new files can be created, even if space is available!
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Real-World Example

A **system administrator** needs to:

1. **Check disk space before installing a new application.**
2. **Find out which partition is nearly full.**
3. **Monitor inode usage to avoid storage issues.**

What did they do?

1. Ran **df -h** to **check available disk space**.
2. Used **df -h /var** to **analyze a specific partition**.
3. Checked **df -i** to **ensure inodes were available**.

Result? They **cleared unnecessary files** and **ensured enough space** for the new installation!

Section 3: Know More

Frequently Asked Questions

1. **How do I check only a specific partition?**

```
df -h /home
```

2. **How do I check disk usage in human-readable format?**

```
df -h
```

3. **What is the difference between **df** and **du**?**
 - **df** shows free and used disk space at the file system level.
 - **du** analyzes space used by individual files and directories.



4. **Why is my disk full but `df` shows free space?**
 - Run `df -i` to check if **inodes** are exhausted.
 5. **Where can I practice disk management?**
 - Use a **Linux Virtual Machine**.
 - Try **monitoring storage** on cloud servers like AWS.
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Conclusion

The `df` command **helps monitor disk space, prevent storage issues, and optimize system performance.**

Start by **checking basic disk usage, using human-readable formats, and analyzing partitions**, and soon you'll be **managing Linux storage like an expert!**